Stable Answers, Unfinished Reasoning: The Stability–Terminality Gap in Probe-Based Early Exit

Anonymous ACL submission

Abstract

Reasoning language models can spend thousands of tokens on a single problem, motivating early-exit methods that stop generation once an answer appears settled. A natural proxy for this state is intermediate-answer *stability*: periodically probe a partial reasoning trajectory and stop when recent answers agree. We ask whether such stability provides reliable evidence that reasoning has *terminated*.

On frozen trajectories from two models and three mathematical reasoning benchmarks, we preregister and evaluate 3,520 stopping rules varying the agreement window, threshold, probe schedule, maturity requirement, answer validity, and reported certainty. None is simultaneously safe and token-saving under any of three predefined operating points. Larger windows reduce accuracy loss only by firing later and less often, causing token savings to collapse.

We trace this trade-off to a mismatch between what probes measure and what early exit requires. Re-probing the same prefix with two wordings produces different answers 57% of the time in the first tenth of a trajectory, compared with 11% near its end. Manual analysis of 134 stopped-but-wrong cases further finds that 60% commit to answers on which the trajectory had not converged, while only 20% reflect settled but incorrect reasoning. A boundary-confidence positive control succeeds under the same evaluation, ruling out the impossibility of early exit itself. Together, these results reveal a *stability–terminality gap*: repeated intermediate-answer agreement alone does not establish that a reasoning trajectory has finished.

1 Introduction

Reasoning language models (DeepSeek-AI, 2025; Qwen Team, 2025; OpenAI, 2024) can spend thousands of tokens on a single answer. *Early exit*—halting a chain of thought once its answer is settled—could cut that cost (Chen et al., 2024; Fu et al., 2024), but only if the stopping signal is safe. A widely used signal is *intermediate answer consensus*: at regular intervals a partial trajectory is probed for its current answer (e.g., by appending a short "`Final Answer:`" suffix) and generation is stopped once recent probes *agree* (Fu et al., 2024). The underlying assumption is simple: if the model repeatedly says x, it has probably finished reasoning about x, and recent work reports it paying off—on several benchmarks answers "converge" after $\sim$60% of the reasoning, and stopping on consensus saves tokens with little *aggregate* accuracy loss (Liu and Wang, 2025; Fu et al., 2024). This paper asks whether that apparent success is *safe*: **is intermediate agreement a signal one can stop on without quietly losing accuracy?**

Our answer is **no**, and the reason is not a badly tuned threshold—it is the signal itself: repeated intermediate agreement measures whether the current answer is *stable*, not whether the reasoning has *terminated*, and the two come apart—a *stability–terminality gap*. This reconciles the apparent success above: the aggregate savings are real, but they ride on a *directional* accuracy tax that safety-blind, probe-cost-free evaluation averages away. Because continued reasoning corrects the answer far more often than it breaks it, stopping on consensus[1] destroys a correct-in-the-end answer up to **45** times more often at an aggressive stop (Section 5)—a recovery asymmetry that is a trajectory-level property, so no threshold in the searched space escapes it. We make the claim hard to dismiss as "you did not tune it well" by searching the space exhaustively under commitment: on frozen trajectories from two models and three benchmarks we run a *preregistered* sweep of the consensus signal—a

[1] "False consensus" here denotes non-terminal intermediate probe agreement, unrelated to the social-psychology term (Ross et al., 1977); we study answer-level, not token/layer-level, early exit (Graves, 2016; Schuster et al., 2022).

window size $\times$ share threshold family plus operational knobs, 3,520 rules—with problem-grouped splits and acceptance gates fixed in advance (Sections 3–4). The outcome does not depend on any single fragile estimate: **not one of the 3,520 rules is both safe and saving—none clears any gate**, and even large windows only trade the drop away for zero saving. To show this is the *signal* and not early exit, we sweep one non-consensus method—DEER, which reads the model's confidence at a reasoning boundary—through the identical pipeline and gates: it clears all three, losing as little as 0.3 pp while saving 28%. The consensus frontier reproduces on a held-out test split (r=0.98) and on unseen models at 4$\times$ the scale and a different architecture (Section 4). Two published-signal checks bracket the contrast: a faithful Certaindex reproduction on the same trajectories collapses by 56–70 pp, while DEER clears the gates on the unseen models too. Figure 1 is the whole paper in one picture: the consensus rules that clear the gates in-sample on train leave the safe-and-saving corner on dev and test, while the three DEER operating points selected on dev stay in it out of sample.

Contribution. Our contribution is to identify and characterize a *stability–terminality gap* in probe-based early exit—intermediate answer agreement measures whether the current answer is *stable under a fixed probing procedure*, not whether the reasoning has *terminated*—and to show, under preregistration, that the gap cannot be tuned away. **(i) The gap**, characterized from two sides: probe reads are wording-unstable early (two wordings disagree ~57% of the time in the first tenth of a trajectory, 11% near its end) and stopped-but-wrong commitments are mostly non-terminal (61% vs. 18% settled-but-wrong), reflecting a directional "continued-reasoning-corrects" asymmetry (Section 5). **(ii) A preregistered, probe-independent sweep** of 3,520 rules (frozen trajectories, gates fixed in advance; released in full) under which *no* consensus rule is both safe and saving, so the gap is not a bad-threshold artifact (Sections 3–4). **(iii) Generalization and a positive control**: the frontier reproduces on a held-out split and two unseen models, a faithful Certaindex reproduction collapses, and a non-consensus signal (DEER) swept identically *clears the same gates*—localizing the failure to the probe-agreement signal rather than to early exit (Sections 4–5).

2 Related Work

Probe-based / consensus early exit. The methods we scrutinize periodically query a partial reasoning trajectory for its current answer and halt on agreement. Dynasor / Certaindex (Fu et al., 2024) formalize this with a "certainty index" over recent probe answers and terminate early when serving reasoning programs, one of several responses to the *overthinking* of reasoning models (Chen et al., 2024; Yang et al., 2025). The underlying assumption is stated most strongly by Liu and Wang (2025), who report that reasoning answers "converge" after only ~60% of steps and that stopping on answer consistency saves tokens with little or no *aggregate* accuracy drop. Learned stoppers read the same probe signal through richer features: LearnStop (Dong et al., 2026) predicts prefix correctness from prefix-observable quantities—answer stability, prefix vote share, confidence, entropy—and finds, as we do, that no single aggressive policy is universally safe (some datasets "admit no certifiable aggressive policy at all"). We adopt the same probe mechanism (a short boxed-answer suffix) and notion of probe agreement, but rather than reporting one aggregate operating point we ask, under *preregistered* safety gates and with probe cost charged, whether *any* rule in this family is both safe and saving; the stability–terminality gap (Section 5) reconciles our empty corner on competition math with those optimistic aggregate reports.

Early-stopping self-consistency and self-correction. Probe-based early exit is the online, *single-trajectory* analogue of early-stopping self-consistency (Wang et al., 2023; Aggarwal et al., 2023; Li et al., 2024): it votes across snapshots of one in-progress trajectory rather than across finished samples. That difference is the crux—snapshots of an unfinished trajectory are not exchangeable draws from a stable answer distribution, because the trajectory is still correcting itself. This intrinsic self-correction differs from the prompted, post-hoc self-correction that Huang et al. (2024) find unreliable: we measure answer evolution *within* one uninterrupted trajectory, and find (Section 4) that a model's mid-trajectory answer is far from final.

Alternative termination signals. Rather than probe agreement, a stop can read other signals: verifier- and process-reward-guided decoding (Lightman et al., 2024; Wang et al., 2024) score

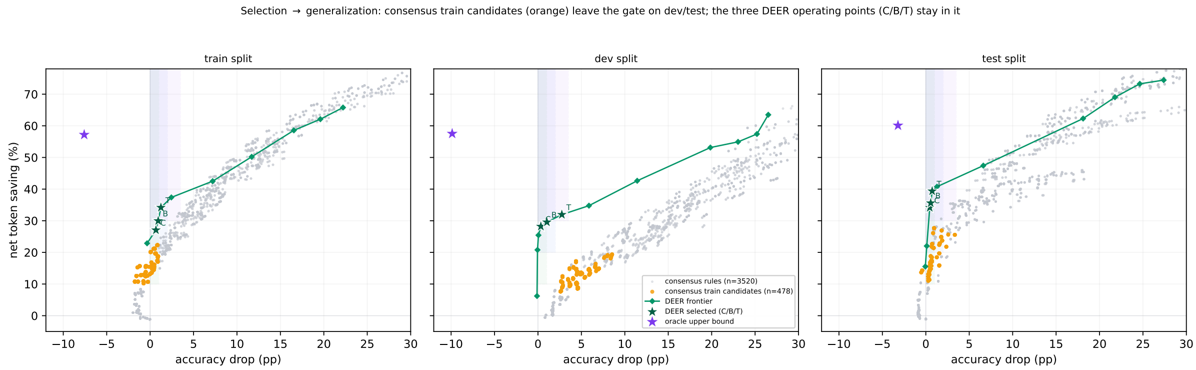

Figure 1: **Selection → generalization, in one picture** (two dev models; macro over environments; net saving vs. accuracy drop). **Grey**: all 3,520 consensus rules. **Orange**: the 478 rules that clear the conservative gate *in-sample on train*; carried unchanged to *dev* and *test* the same rules fall outside the gate region (higher drop, lower saving), so no consensus rule holds up out of sample. **Green**: the DEER threshold frontier (identical pipeline and accounting); labelled stars are the three dev-selected operating points (**C/B/T**), which reproduce inside the gate out of sample (§4.5). **Purple star**: an oracle upper bound (earliest correct probe), far in the corner neither method reaches. The failure is a property of the consensus *signal*, not of early exit.

partial reasoning, and adaptive-computation methods learn a halting signal at the token or layer level (Graves, 2016; Schuster et al., 2022). DEER (Yang et al., 2025) stops on the model's confidence in a trial answer at a reasoning boundary, and CoDE-Stop (Hosseini et al., 2026) tracks how that confidence *evolves* rather than whether probes agree—both read a forward-looking signal, not consensus. We use DEER only as a *positive control* (Section 5): run through the identical pipeline and acceptance gates, it clears them where no consensus rule does, indicating that the failure is specific to the probe-agreement signal rather than to early exit itself. We do not claim this broader class of signals is categorically superior—our experiments isolate one signal, not a taxonomy of methods. Test-time scaling trades compute for accuracy via repeated sampling and search (Snell et al., 2024; Brown et al., 2024); early exit is its dual, spending less without losing accuracy.

Preregistration. To keep an exhaustive rule search from silently overfitting to a favorable operating point, we fix the rule schema, splits, and acceptance gates before looking at held-out data (Nosek et al., 2025). This preregistration (Section 3) is what makes a *negative* result interpretable.

3 Method

To ask whether *any* probe-based rule can succeed, we need (i) an evaluation that a rule cannot game by changing the reasoning it observes, and (ii) a selection procedure that cannot silently overfit to a lucky operating point. We preregister both.

3.1 Probe-independent, frozen trajectories

For each problem we generate exactly *one* main trajectory in a single request (caps: 16K tokens for MATH500 and AMC23, 32K for AIME24) and freeze it. All probing is then performed *offline* against fixed text prefixes of that frozen trajectory. Because probes never re-enter the decode, changing a probe schedule—or a stopping rule—cannot change the underlying reasoning; every rule is scored against the same trajectories. We build two offline probe banks: a *dense* bank that re-probes every 64 tokens with simple@32 (a 32-sample boxed-answer probe), and an *adaptive* bank that additionally fires event-triggered probes at entropy drops and at conclusion/reflection/answer markers.

3.2 Token accounting

For a rule that stops a problem at token position s after consuming probe output p, we charge total decode tokens $T = s + p$ and compare against the frozen baseline B (the full trajectory, capped at budget). *Net* savings is $(B - T)/B$ and *gross* savings is $(B - s)/B$; their gap is the probe cost. The accuracy drop compares the correctness of the committed answer at s against that of the frozen final answer (it is not an agreement rate between them). Two quantities therefore move independently, and we track both: *where* the rule stops (setting the accuracy drop and the main-token saving s) and *how much it probes* to get there (adding p). Section 5 shows that only the first is intrinsic.

3.3 The consensus signal: window size and share threshold

The consensus signal any probe-based rule can read reduces to two hyperparameters: a **window size** W (how many of the most recent probes to inspect) and a **share threshold** s (what fraction of that window must agree on a single answer before a stop is allowed). This pair spans the family to a close approximation: $W{=}1$ is the "latest-probe" rule, $s{=}1.0$ is strict unanimity over the last W probes (the Certaindex-style stop), and an entropy-over-the-window criterion closely tracks the same agreement and adds no materially new operating point on our grid. We sweep $W \in \{1, 3, 5, 8, 12, 16, 24, 30\}$ and $s \in \{0.6, 0.8, 1.0\}$, deliberately including *large* windows so a rule can demand a long, stable agreement before committing.

Around this core we vary only the operational knobs that govern *when* and *how* the signal is read: **probe schedule** (fixed interval $\in \{64, 128, 256, 512\}$ tokens, or event-triggered probing at conclusion/entropy/reflection markers); **maturity** (a minimum-token floor $\in \{0, 512, 1024, 2048, 4096\}$ before any stop is allowed); **validity** (an answer-shape filter, rejecting empty/letter answers or only empties); and **certainty** (optionally requiring the supporting probes to be flagged certain). Enumerating this preregistered grid over a fixed-schedule and an event-triggered family yields 3,520 candidate consensus rules; formal definitions are in Appendix A.

A better signal, swept identically. To test whether the failure is early exit *itself* or the consensus *signal*, we sweep one non-consensus method through the same pipeline, gates, and token accounting. **DEER** (Yang et al., 2025) stops on the model's own confidence in a trial answer at a reasoning boundary; its only hyperparameter is a confidence threshold, which we sweep over 14 values. We use the stronger *trial-answer-submit* variant (commit the trial answer directly, no separate readout) rather than the faithful readout reproduction—our aim here is not to reproduce prior work but to ask whether *any* rule can clear the gates. Placing DEER and consensus on one frontier under identical accounting is what makes the comparison a controlled test of the signal.

Operating point	Max total acc. drop	Min total saving	psf floor
conservative	1.0 pp	10%	0.80
balanced	2.0 pp	20%	0.80
token_efficient	3.5 pp	30%	0.70

Table 1: Preregistered acceptance gates. A rule must keep the macro-averaged accuracy drop under "max total acc. drop," deliver at least "min total saving" net tokens, and post positive net saving on at least a psf fraction of environments. Gates are fixed in advance and *not* relaxed post hoc.

3.4 Models, benchmarks, splits

Development uses two models—DeepSeek-R1-Distill-Qwen-7B and Qwen3-8B—on three benchmarks (MATH500, AMC23, AIME24), each at seeds 42/43/44: 18 model$\times$benchmark$\times$seed environments. Problems are partitioned *by problem id* into 60/20/20 train/dev/test splits (MATH500 300/100/100, AMC23 24/8/8, AIME24 18/6/6), so no problem leaks across splits. Confirmation reserves seeds 45/46/47 plus two unseen models—DeepSeek-R1-Distill-Llama-8B and Distill-Qwen-32B—all evaluated on the *test* split only.

3.5 Preregistered operating points and gates

We fix three operating points and their acceptance gates *before* looking at held-out data (Table 1). Each gate is applied in order: first the **total accuracy drop** (macro-averaged over the 18 environments) must stay under a cap; then the **total net token saving** (macro-averaged) must clear a floor; and finally a **positive-saving fraction** (psf, the fraction of environments with strictly positive net saving) floor must be met. A rule is accepted at an operating point only if it satisfies all three on train+dev. The saving floor is what distinguishes a genuinely useful early exit from a rule that stays accurate only by almost never stopping.

3.6 Metric resolution

The total accuracy drop is a macro-average over the 18 environments, each model$\times$benchmark$\times$seed weighted equally, so selection rewards a rule robust across benchmarks, models, and randomness rather than one that overfits a single setting. Pooling problems instead would let MATH500 (100 dev problems per seed vs. 8/6 for AMC23/AIME24) dominate, so a MATH500-only winner could pass; equal per-environment weight forces a rule to hold on the harder competition sets and keeps any single problem from swinging the result (one AIME24 problem is 16.7 pp of its split but only $\sim 1/108$ of the macro drop). Gates are read on dev (train reported alongside); because no rule clears any gate by a wide margin (Section 4), the finding does not rest on a fragile cell, and the test split plus two unseen models provide the out-of-distribution check.

3.7 Preregistration commitments

Three commitments make the result interpretable. **(C1)** The test split and both confirmation models are never read during rule selection. **(C2)** All three operating points and their gates in Table 1 are fixed before seeing held-out data and are not relaxed afterwards; if no rule passes a gate, that is reported as the finding rather than engineered away, and all three points are reported regardless of outcome. **(C3)** The frozen rule set is fixed on train+dev and its hash recorded before any test/confirmation evaluation. Section 4 reports what happened when we applied C2.

4 Experiments

We first confirm the phenomenon the paper turns on—intermediate consensus is non-terminal (§4.1)—then report the preregistered sweep and its central negative result: *no* consensus rule is both safe and saving (§4.2). We then show the failure is specific to the *signal*, not to early exit, by sweeping a non-consensus method (DEER) through the identical pipeline and gates and finding that it *does* clear them (§4.3); a faithful Certaindex reproduction on the same trajectories collapses (§4.4). Finally we confirm the frontier out of distribution (§4.5).

4.1 False consensus is real

We log a dense probe stream for every MATH500 problem under DeepSeek-R1-Distill-Qwen-7B at the main 16K/32K budgets (all 500 problems, three seeds; 1,500 trajectories), never intervening—the probe is a separate forward pass that only observes the main trajectory (full setup in Appendix E)—and read off four facts.

(1) Final agreement is reliable; early agreement is not. By the time a trajectory finishes, agreement does track correctness: *cumulative* agreement across all probes is 97.8% correct (186 trajectories reach it), and even the final five-probe *window* is 90.4% correct (1,205). But an online rule cannot wait for the end—it fires the *first* time probes agree, mid-trajectory, where the answer is far from settled. The operative question is not whether *final* consensus is trustworthy but whether *early* consensus is—and it is not (facts 2–4). **(2) Recovery is common.** Of the 1,148 trajectories whose *first* probe is wrong, 89.1% eventually flip to correct; of the 736 that formed a three-probe consensus *different* from their final answer, 84.2% ended up correct. **(3) Later consensus is worse.** Consensus formed before 512 tokens is 91.0% correct, but consensus forming only after 2048 tokens is 71.6% correct—hard problems converge slowly *and* to worse answers. **(4) The cost lands at the stop.** A naive stop (first three consecutive agreeing, certain, non-empty probes) fires on $1{,}477/1{,}500$ trajectories and commits to answers correct only 50.5% of the time, whereas letting those same problems run to completion reaches 90.7%: a 40.2 pp loss.[2]

A two-annotator study of stopped-but-wrong cases—on a separate exploratory dense-logging pass (Appendix E.5)—confirms the failure is one of *timing*, not merely difficulty: 61% commit to an answer on which the trajectory had *not yet converged*, against only 18% that are settled but wrong. Intermediate *stability* is thus not evidence of *terminality*.

A good stopping rule must therefore be much more than "agreement." The rest of this section asks whether the *best possible* such rule, drawn from a large preregistered space, can be both safe and economical.

4.2 The safe-and-saving corner is empty for consensus

We replay all 3,520 consensus rules on the frozen trajectories and aggregate to per-environment metrics—126,720 rows $=$ 3,520 rules $\times$ 18 environments $\times$ 2 splits (train, dev)—then, per rule, its total (macro) accuracy drop and net token saving (accounting of Section 3.2). Applying the preregistered gates yields the paper's central negative result.

The outcome holds across environments: across all 126,720 rule–environment evaluations accuracy strictly *falls* in 64.9% and rises in only 10.5% (mean drop 13.0 pp; over rules, median dev drop 13.2 pp, minimum 0.81 pp). **Not one of the** 3,520

[2] A simplified heuristic, *not* the Certaindex algorithm (§4.4); it fires early because the first stable, certain agreement is typically a mid-trajectory placeholder. The swept rules below and the mechanism of Section 5 quantify the same directional accuracy tax across the full rule space.

Method	Acc.	Δacc (macro)	Net saving
Full generation	82.5%	—	0%
Consensus (drop$\leq$ 1.0)	81.6%	-0.9 pp	$+0.2\%$
Consensus (save$\geq$ 10%)	79.9%	-2.7 pp	$+10.7\%$
DEER (conservative)	82.5%	-0.33 pp	$+28.2\%$
DEER (balanced)	81.9%	-1.03 pp	$+29.6\%$
DEER (token_eff.)	80.1%	-2.75 pp	$+31.9\%$

Table 2: Dev operating points (macro over 18 environments). No consensus rule is simultaneously safe (≤ 1.0 pp) and saving ($\geq 10\%$): capping the drop forces saving to near zero, and demanding saving forces a large drop. DEER, reading a non-consensus signal, clears all three gates on the same trajectories under the same accounting.

rules clears any gate. The reason is the shape of the frontier (dev panel of Figure 1): staying accurate and saving tokens are in direct tension. Only 40 rules keep the total drop at or below the conservative 1.0 pp cap, and the most any of them saves is 0.2%—far under the 10% floor; conversely, the first rule to save 10% already costs 2.66 pp, saving 20% costs 6.17 pp, and saving 30% costs 11.8 pp. Crucially, the sweep now includes *large* windows (up to W=30): these do buy low accuracy drop, but only by stopping so late that net saving collapses to zero. No setting of the consensus signal escapes the trade-off; the safe-and-saving corner is empty.

4.3 Early exit *is* possible—with a different signal

The same gates are cleared, comfortably, by a rule that does *not* read probe agreement. Sweeping DEER's confidence threshold through the identical pipeline and token accounting (Section 3.3), its trial-answer-submit variant passes all three operating points on dev (conservative -0.33 pp/28.2%, balanced -1.03 pp/29.6%, token-efficient -2.75 pp/31.9%; accuracy-neutral -0.06 pp at 20.8% under a stricter threshold; Table 2). In the dev panel of Figure 1 its frontier sits *above* the consensus frontier everywhere in the safe region. Early exit is therefore not impossible; the consensus *signal* is what cannot make it safe.

4.4 Consensus in the wild: a faithful Certaindex reproduction collapses

The sweep shows no *tuned* consensus rule is safe; a published one shows the same failure at its shipped operating point. In a common harness (same frozen trajectories, simple@32 probe bank, and token accounting) we reproduce three prior stoppers—**CertaIndex** (Fu et al., 2024), the faithful share-threshold stop and direct descendant of the consensus family; **DEER** (Yang et al., 2025), here the *faithful* readout (the stronger trial-answer-submit variant is what we swept in §4.3); and token-level **TJE** (Anonymous, 2026) (Table 5, appendix). CertaIndex, at a 99.8% stop rate and 70 pp lost, is the accuracy tax paid in full—landing exactly where the swept frontier says it must—and TJE is unsafe on the weaker model, whereas even the faithful DEER readout, a non-consensus signal, avoids collapse. That only the non-consensus signal survives is the crux we analyze in Section 5.

4.5 The frontier reproduces out of distribution

We read the test split once (commitment C3). Because the development result is that *no* consensus rule clears any gate, there is nothing to confirm by re-selection; we ask whether the *frontier* reproduces, ranking rules on dev and measuring on test.

Held-out test split. Each consensus rule's total drop on test tracks its dev value closely ($r = 0.98$). The split-invariant claim is the empty *joint* gate: **no consensus rule clears the conservative gate on both splits**—0 on dev, and the 444 rules admissible on *test alone* are exactly the in-sample winners the held-out split exists to reject (all fail on dev). DEER, by contrast, remains admissible across splits: 3 of its thresholds clear the conservative gate on *both* dev and test (and 5 clear balanced on both), so the cross-signal contrast is not a dev artifact.

Unseen scale (4×) and architecture. Both unseen models are evaluated on the test split at three seeds (45/46/47; nine environments each). The consensus frontier reproduces against dev drop on Distill-Qwen-32B ($r = 0.97$) and on the held-out architecture DeepSeek-R1-Distill-Llama-8B ($r = 0.94$).[3] The *conservative* gate stays empty on every model (the best rule under a 1.0 pp drop saves 0.6% on 32B, 9.3% on Llama). A mild scale effect appears at looser gates—the 4× larger 32B admits a few rules in-sample (4 balanced, 6 token-efficient) as its intermediate answers stabilize earlier—but these are not dev-selected (dev admits none), and

[3] Llama-8B was re-collected after fixing a chat-template defect (a missing beginning-of-sequence token); the corrected model scores ∼88% on MATH500.

Llama stays empty (0/0/0). DEER, swept identically, **clears the gates on both**: accuracy-neutral on Qwen-32B (-0.24 pp at 32.4%, τ=0.97) and 0.67 pp at 26.7% on Llama-8B (τ=0.99). Consensus is unsafe, and DEER safe-and-saving, across all four axes—benchmark, seed, 4× scale, and architecture.

Figure 2 shows this per model: tracking the *same* train-selected consensus rules (orange) and dev-selected DEER points (stars) onto each test panel, the orange candidates scatter to low saving or higher drop—they do not transfer—while the DEER points land back inside the gate (per-benchmark panels in Figure 3, appendix; on the small AIME24 set the frontiers are noisier and intertwine). Each panel also plots an *oracle* upper bound (earliest correct probe), which sits far in the safe-and-saving corner (2–5 pp negative drop at 40–80% saving) that neither method reaches—headroom a better *realizable* signal might recover.

5 Analysis

Section 4 is an existence proof over 3,520 rules, two seen and two unseen models, and both splits. This section explains *why* it holds and why it generalizes beyond the exact rules we searched. The one-line reason is a *stability–terminality gap*: probe agreement measures whether the current answer is *stable* under a fixed probing procedure, not whether the reasoning has *terminated*. Underlying it is a directional property of the trajectory itself—continued reasoning keeps correcting the answer—that no threshold on the agreement signal removes.

5.1 The recovery asymmetry is trajectory-level

Count, among stops that *change* correctness relative to the trajectory's own final answer, a correct recovery destroyed (final-correct, stop-wrong) versus a wrong answer luckily banked (final-wrong, stop-correct). Sampling noise predicts $\approx$1:1; a "continued reasoning corrects" effect predicts $\gg 1$. On the frozen replay this ratio is window-dependent—$\approx$45:1 at a latest-probe stop (W=1) down to $\approx$2:1 at W=30—but the low end is bought only by stopping so rarely that net saving vanishes (W=30 fires on 121 dev problems vs. 668 at W=1). At any window that saves tokens the ratio stays well above 1:1: consensus destroys a correct-in-the-end answer many times more often than it banks a wrong one—the recovery phenomenon of §4.1 (84.2%, 89.1%) as a signed effect at the main budgets. The asymmetry is trajectory-level: a rule stopping at position s forgoes whatever correction the trajectory would make after s, so wherever a consensus rule fires early it sits in a region where continued reasoning still corrects. The *size* of the realized accuracy cost depends on the probe schedule and stopping rule—what the sweep varies—but across the grid (64/128/256/512 intervals, 8 windows, 3 thresholds) it never reaches zero: no threshold on the agreement signal escapes the region where reasoning is still moving. Agreement thus reports only what the model has *said recently*—not whether reasoning has *terminated*. DEER, by contrast, holds $\approx$ 2.4–3.5:1 *while* saving 28–32%, committing mainly once the trajectory has settled.

5.2 The probe reads stability, not terminality

Two direct measurements make the gap concrete from opposite sides. **Probe reads are wording-unstable early.** We re-probe the *same* frozen prefixes with two differently worded answer queries—the simple@32 suffix and a Certaindex-style suffix—and measure how often they extract the *same* current answer (DeepSeek-R1-Distill-Qwen-7B, MATH500; 2,898 probe points binned by trajectory position; Table 7). Agreement rises monotonically: the two wordings disagree on the current answer ∼57% of the time early in a trajectory (only 41% agree in the first 5%), but just 11% near its end (89% agree). Early on, the "answer" a probe surfaces is an artifact of how it is asked as much as of the underlying reasoning—the same prefix probed two ways yields two answers—and the wordings converge only as the trajectory settles. Agreement under one fixed wording therefore shows only that the model *repeats* an answer under that procedure—not that the answer is robust to how it is asked, and not that reasoning has finished. **Stopped-but-wrong answers are mostly non-terminal.** A two-annotator study of stopped-but-wrong cases (§4.1; Appendix E.5) shows the same gap from the error side: 61% of stopped-but-wrong commitments are to answers the trajectory had *not yet converged* on (type D), versus only ∼18% that are settled but wrong (type A). Both say the same thing: probe agreement is a stability signal, and stability is reached well before terminality.

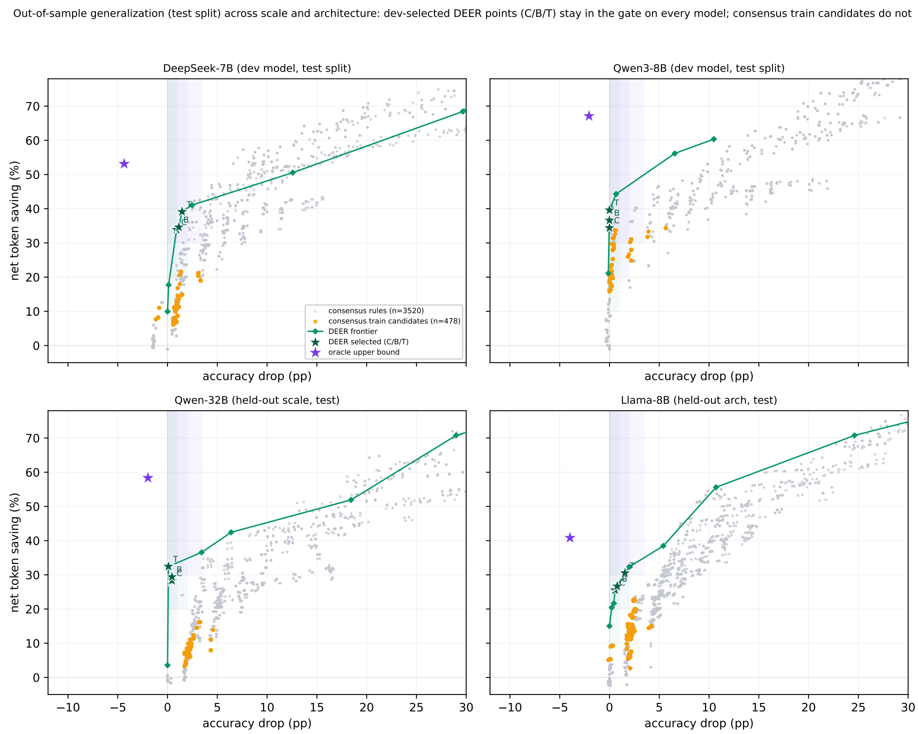

Figure 2: **Out-of-sample generalization across scale and architecture** (all panels test split). Grey: consensus rules; **orange**: consensus train candidates; **green**: DEER frontier with the three dev-selected operating points (C/B/T); **purple**: oracle upper bound. The two development models (top) are shown on their held-out test seeds; the two held-out models (bottom) are entirely unseen. The dev-selected DEER points stay inside the conservative gate on every model; no consensus rule does.

5.3 It is the signal—not dense probing, not early exit

Two objections remain; dismissing both localizes the failure to the signal. *"It is just dense probing."* Besides the accuracy tax, a rule pays a *probe tax*: the probe output p (Section 3.2). Under dense re-probing a late-stopping large-window rule pays p over a long prefix, opening a gap between its *gross* and *net* savings (Table 6). But the probe tax scales with probe density and is reducible (sparser or shorter probes, KV-cache reuse), whereas the accuracy floor is probe-independent; it is the accuracy floor—which DEER escapes—not the probe tax, that keeps the corner empty. *"Maybe early exit itself is impossible."* It is not. Every method that stops on *probe agreement*—the swept consensus family and CertaIndex alike—either fails a gate or collapses (CertaIndex loses 70 pp at a 99.8% stop rate, the accuracy tax paid in full). The one method that clears the gates, DEER, does not read probe agreement at all: it stops on the model's confidence in a trial answer, and in Figure 1 its frontier sits *above* the consensus Pareto boundary throughout the safe region. Swept through the identical pipeline, gates, and accounting, this is a controlled test: the same machinery succeeds on DEER's boundary-confidence signal and fails on consensus. The failure is a property of the probe-agreement *signal*, not of early exit.

6 Conclusion

Intermediate answer-consensus is not a safe early-exit signal for reasoning LLMs, and not because the rule is untuned: agreement measures answer stability under a fixed probe, not whether reasoning has terminated—a *stability–terminality gap*. Because continued reasoning corrects the answer far more often than it breaks it, stopping on consensus destroys a correct-in-the-end answer up to 45 times more often than it banks one. This trajectory-level recovery asymmetry leaves the safe-and-saving corner empty across all 3,520 rules, reproduced on a held-out split and two unseen models, whereas a non-consensus signal (DEER) swept identically clears every gate, localizing the failure to the *signal*. Probe agreement *alone* does not establish terminality and cannot be tuned into a safe stop; which signal *should* replace it we leave open.

Limitations

Scope of the negative result. The central result—no consensus rule clears any gate—is established on 18 development environments (two models, three benchmarks, seeds $42/43/44$) and confirmed on the held-out test split and two unseen models (Section 4). It is a statement about the space searched: one boxed-answer probe suffix (simple@32) and one preregistered consensus schema (3,520 rules over a window-size $\times$ share-threshold family plus operational knobs). We separately probe robustness to the answer-query *wording*—comparing a second wording position-by-position (Section 5.2)—but do not claim to have exhausted every possible wording, and non-consensus signals beyond our DEER control are not swept; the mechanism (Section 5) is our argument that consensus variants would not help, but it is an argument, not an exhaustive proof.

DEER as the positive control, not a benchmarked SOTA. We sweep DEER's trial-answer-submit variant to establish that *some* signal clears the gates the consensus family fails (on the development models, the held-out test split, and both unseen models); we do not claim it is the best possible early-exit method. Its role is a controlled positive control for the signal, not a leaderboard entry.

Small held-out sets on hard benchmarks. AIME24 and AMC23 are small (dev splits of 6 and 8 problems per seed), so those per-environment cells are noisy. We damp this with three seeds per environment and macro-averaging over the 18 development environments, and read the unseen-model results (also three test seeds each, $45/46/47$) as frontier-reproduction evidence rather than as independent gate tests (Section 3.6).

Probe-tax dependence of the savings axis. Net-savings numbers charge dense re-probing (every 64 tokens, 32-token probes); a sparser or KV-cache-reusing schedule would change the *savings* axis and could move some rules to positive net savings. We separate gross from net savings (Section 5) precisely so the probe-tax-dependent part is distinguishable; the accuracy-tax result does not depend on the schedule.

Domain. All benchmarks are competition mathematics with checkable final answers. Whether false consensus and the accuracy tax transfer to open-ended reasoning, code, or agentic tasks is out of scope.

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A Rule Schema Details

The consensus schema of Section 3 centers on two hyperparameters and a few operational knobs, enumerating to 3,520 candidate rules over a fixed-schedule and an event-triggered family. Each rule fixes:

1. **Window size** $W \in \{1, 3, 5, 8, 12, 16, 24, 30\}$: the number of most recent probes inspected. $W{=}1$ is the latest-probe rule.
2. **Share threshold** $s \in \{0.6, 0.8, 1.0\}$: the fraction of the last W probes that must agree on one answer (share measured over the full window, so an empty or dissenting probe counts against consensus); $s{=}1.0$ is strict unanimity. An entropy-over-the-window criterion closely tracks the same agreement and adds no materially new operating point on our grid, so it is not swept separately.
3. **Probe schedule**: fixed interval $\in \{64, 128, 256, 512\}$ tokens, or `adaptive_event` with entropy-drop and marker triggers and a fallback interval.
4. **Maturity**: a minimum-token floor $\in \{0, 512, 1024, 2048, 4096\}$ before any stop is allowed.
5. **Certainty**: optionally require the supporting probes flagged certain (minimum fraction).
6. **Validity**: answer-shape filter (reject empty / single-letter answers to suppress the Type-E probe artifact of Appendix E), or accept any non-empty answer.

Certainty. A probe is *certain* when its boxed answer is emitted with high token-level confidence: concretely, the mean probability of the answer tokens under the probe forward pass exceeds a fixed threshold (the Certaindex certainty flag). For the simple@32 bank this is aggregated over the 32 samples as the fraction agreeing on the modal answer. Dimension 5 gates a stop on the fraction of the supporting probes that are certain.

Entropy triggers. The `adaptive_event` schedule fires an extra probe when the token-level predictive entropy of the main decode drops by at least ΔH relative to a trailing baseline (an "the model just committed" signal), and at conclusion/reflection/answer marker tokens (e.g., "therefore", "the answer is"), subject to a minimum inter-probe gap and a fallback interval.

B Preregistration Text

The protocol version, split manifest hash, candidate-rule generation script, and the three operating-point gates (Table 1) were fixed before

Window W	Min total drop (pp)	Max net saving
1	22.25	98.1%
3	11.76	93.8%
5	8.01	87.9%
8	4.91	72.9%
12	1.85	62.0%
16	1.74	55.1%
24	1.69	43.2%
30	0.81	38.8%

Table 3: Consensus frontier along the window axis on dev (over all share thresholds and operational knobs at each W). Larger windows buy a smaller minimum accuracy drop but only by stopping later, so the maximum net saving falls in lockstep; the two never meet in the gate region.

Threshold τ	Total drop (pp)	Net saving
0.9999	−0.06	20.8%
0.999	0.06	25.4%
0.995	0.33	28.2%
0.99	1.03	29.6%
0.97	2.75	31.9%
0.95	5.87	34.8%
0.90	11.43	42.6%

Table 4: DEER trial-answer-submit threshold frontier on dev (macro over 18 environments, same token accounting). Unlike consensus, low drop coincides with substantial saving—the conservative/balanced/token-efficient gates are all cleared around $\tau \in [0.97, 0.995]$.

any held-out evaluation. Commitments C1–C3 (Section 3) were recorded with the protocol. All three operating-point gates were fixed in advance; when the conservative gate proved empty we followed C2 and report the empty gate as the finding rather than relaxing it, and all three points are reported for both consensus and DEER regardless of outcome.

C Reproducibility

Frozen trajectories, offline probe banks, the sweep archive (126,720 dev metric rows plus the DEER threshold sweep), and the aggregation scripts are released. Tables 3–4 were reproduced to the reported precision directly from the released sweep archive. Grading uses a robust answer-equivalence check that tries the raw and normalized reference forms and both string and math-expression equality; per-problem correctness for the frozen baseline is recomputed at replay time rather than trusted from collection.

D Window-Size and DEER Frontiers

Table 3 traces the consensus trade-off directly along the window axis: as the window W grows, the minimum attainable accuracy drop falls monotonically (from 22 pp at W=1 to 0.8 pp at W=30), but the maximum net saving falls with it (from 98% to 39%)—the low-drop region is reachable only at large windows, where saving has already collapsed. This is why no single (W, s) setting lands in the safe-and-saving corner. Table 4 gives the DEER threshold frontier that does reach it.

D.1 Per-benchmark generalization

Figure 3 breaks the out-of-sample result (Section 4.5) down by benchmark, complementing the per-model view in Figure 2.

E False Consensus: Full Analysis

Section 4.1 summarizes, in four facts, what this appendix establishes in full: in an intervention-free setting, intermediate probe consensus is not a reliable stopping signal.

E.1 Setup

We analyze the frozen main trajectories (Section 3): DeepSeek-R1-Distill-Qwen-7B on all 500 MATH500 problems (Lightman et al., 2024; Hendrycks et al., 2021) at the main 16K-token budget, temperature 0.6, top-p 0.95, across three seeds (1,500 trajectories in total).[4] The dense probe bank appends the boxed-answer probe suffix every 64 generated tokens and reads off the model's current answer. The probe is a separate forward pass: the controller only *observes*, never halting or altering the main trajectory. This yields, per trajectory, the full answer sequence a stopping rule would have access to at the budget used throughout the paper.

E.2 Final agreement is reliable; early agreement is not

Agreement measured at the *end* of a trajectory does track correctness. On the 186 trajectories that reach unanimous *cumulative* agreement (all non-empty probes concur), 97.8% are correct—only four whole-trajectory false consensuses; and of the

[4] Train/dev problem ids are collected at seeds 42/43/44 and the held-out test ids at seeds 45/46/47; their union covers all 500 problems. This is a descriptive characterization of the frozen trajectories—it selects and tunes nothing—so it does not touch the preregistered sweep or the test-split commitment.

Method	Model	Accuracy	Δacc	Fair token saving	Stop rate	Signal
Full generation	Qwen3-8B	85.4%	—	0%	0%	—
Full generation	DeepSeek-R1-Distill-Qwen-7B	79.8%	—	0%	0%	—
CertaIndex	Qwen3-8B	15.3%	-70.1 pp	90.1%	99.8%	consensus
CertaIndex	DeepSeek-R1-Distill-Qwen-7B	23.9%	-55.9 pp	76.7%	98.7%	consensus
TJE	Qwen3-8B	85.0%	-0.4 pp	2.0%	22.5%	token-level
TJE	DeepSeek-R1-Distill-Qwen-7B	60.7%	-19.1 pp	65.0%	93.4%	token-level
DEER	Qwen3-8B	86.2%	**+0.8** pp	16.3%	41.4%	boundary conf.
DEER	DeepSeek-R1-Distill-Qwen-7B	74.9%	-4.8 pp	20.2%	56.1%	boundary conf.

Table 5: Prior early-stoppers reproduced on our frozen trajectories (dev, macro over three benchmarks; probe tokens charged; Section 4.4). CertaIndex—the consensus descendant—**collapses** (-56 to -70 pp) while saving 77–90% of tokens: it fires on almost every problem, very early, exactly the false-consensus trap of §4.1. **DEER**, which reads a *different* signal (boundary confidence), is the only method that stays near full accuracy—+0.8 pp on Qwen3-8B—while still saving tokens.

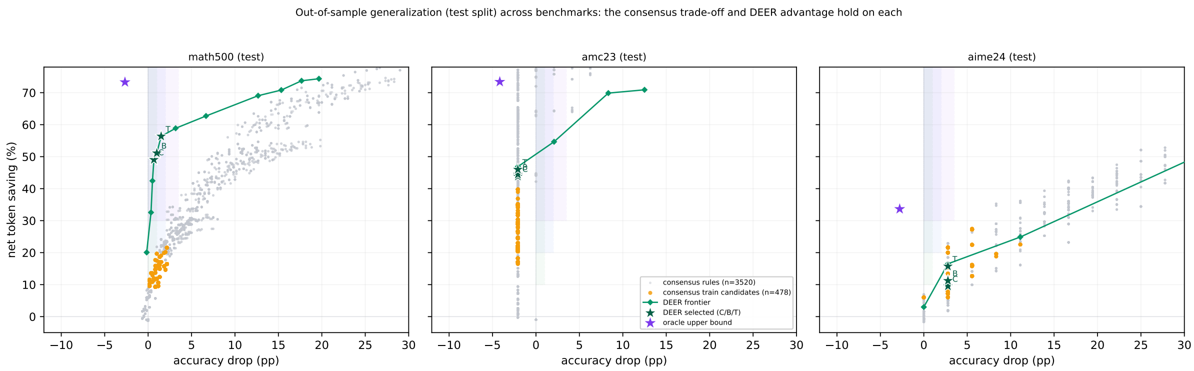

Figure 3: **Out-of-sample generalization across benchmarks** (test split, development models). The consensus trade-off and the DEER advantage hold on MATH500 and AMC23; on the small AIME24 set (few problems per environment) the two frontiers are noisier and intertwine. Markers as in Figure 2.

Consensus point	Gross saving $(B-s)/B$	Net saving $(B-s-p)/B$
save$\geq$ 10% (drop 2.7 pp)	+14.7%	+10.7%
save$\geq$ 20% (drop 6.2 pp)	+27.0%	+20.2%

Table 6: Gross vs. net savings for two consensus frontier points (Section 5.3); the gap ($\sim$4–7 pp) is the probe tax. It shifts the *savings* axis but never rescues the accuracy drop—the intrinsic tax that keeps the corner empty.

1,205 whose *final five-probe window* is unanimous, 90.4% are correct. But this end-state agreement is not what an online stopping rule can act on: a rule fires the *first* time it sees agreement, mid-trajectory, long before the answer has settled. The rest of this appendix shows that *early* agreement—the signal a controller must actually use—is far from terminal.

E.3 The cost lands at the stopping decision

We simulate a *naive consecutive-agreement* stop: halt at the first point where three consecutive probes agree, are non-empty, and are *certain* (the model emits the boxed answer with high token-level confidence; formal definition in Appendix A). This is a simplified heuristic, *not* the Certaindex algorithm, which we reproduce faithfully in §4.4. It fires on 1,477/1,500 trajectories. The committed answers are correct only 50.5% of the time, whereas letting those same problems run to completion reaches 90.7%—a 40.2 **pp accuracy loss** in exchange for an average saving of 3301 tokens, with 731 trajectories stopped on an outright wrong answer.

E.4 Recovery is common, and early stopping kills it

The loss is driven by *recovery*: continued reasoning overturns the intermediate answer far more often than intuition suggests. Of the 736 trajectories that at some point formed a three-probe consensus *different* from their final answer, 620 (84.2%) ended up correct; and of the 1,148 whose *first* probe was wrong, 89.1% eventually flipped to correct. Neither "first impression" nor "local consensus" is close to terminal.

Counterintuitively, *later* consensus is *less* trustworthy: consensus formed before 512 tokens is 91.0% correct, but consensus that only forms after 2048 tokens is 71.6% correct. Hard problems both converge slowly and converge to worse answers—exactly the problems on which an aggressive stopping rule does most damage.

E.5 What the stopped-but-wrong answers are: a two-annotator study

The counts above say stopping is costly; hand analysis says *why*. On the exploratory dense-logging pass (DeepSeek-R1-Distill-Qwen-7B, MATH500, 3072-token budget; the same pass a preliminary 28-case labeling drew from, which this supersedes), we take *all* 134 stopped-but-wrong cases—every problem whose naive-stop answer is wrong—and have two annotators independently label each, shown the problem, the full probe-answer sequence, and the trajectory's final answer. The five-way scheme is **(A)** numeric collapse (stable convergence to a wrong number), **(B)** expression collapse, **(C)** sign error, **(D)** reasoning gap—an intermediate quantity, placeholder, or answer emitted *before the trajectory has converged*—and **(E)** format/option hallucination (a non-multiple-choice problem stably emitting a letter such as “B,” an artifact of the probe mechanism rather than a model belief). The interface and guidelines are released.

Before adjudication the annotators agree on 47.8% of cases (Cohen's κ=0.29, fair); the 70 disagreements are resolved to labels of record under a preregistered tie-break (61 A/D conflicts to D, 9 case-by-case, none left unresolved). The adjudicated distribution is dominated by *non-terminal* errors: **D** = 61.2% (82/134), A = 17.9% (24/134), E = 18.7% (25/134), with B (1.5%) and C (0.7%) negligible. This makes the stability–terminality gap concrete: about 61% of stopped-but-wrong commitments are answers on which the trajectory had *not yet converged* (type D), versus only $\sim$18% that are settled-but-incorrect reasoning (type A)—consensus fires on a mid-trajectory placeholder roughly three times more often than on genuinely finished-but-wrong reasoning. Type E is a methodological finding in its own right: roughly one in five false consensuses is a probe formatting artifact, so any deployed rule should filter answer *shape*.

Position (% traj.)	n	Wordings agree	Probe correct
0–5	34	41.2%	17.6%
5–10	179	47.5%	24.0%
10–15	161	59.6%	28.6%
15–20	150	52.7%	37.3%
20–30	340	60.6%	43.8%
30–40	316	69.6%	56.6%
40–50	325	82.2%	70.8%
50–60	321	87.9%	76.0%
60–70	299	87.6%	79.9%
70–85	422	90.3%	81.8%
85–100	351	88.6%	80.9%
Total	2,898	76.0%	—

Table 7: Two-wording probe agreement by trajectory position (DeepSeek-R1-Distill-Qwen-7B, MATH500, seed 42; simple@32 vs. Certaindex-style suffix). The two wordings extract the same current answer only 41% of the time in the first 5% of a trajectory, rising to $\sim$90% late, so early agreement reflects how the probe is worded, not that reasoning has terminated.

E.6 Probe wording (in)stability: the two-wording measurement

Table 7 gives the raw two-wording agreement referenced in Section 5.2. For every dense probe point on the frozen DeepSeek-R1-Distill-Qwen-7B MATH500 trajectories (seed 42; 2,898 probe points) we extract the current answer with two differently worded queries—the simple@32 boxed-answer suffix and a Certaindex-style suffix—and record whether they agree, binned by the probe's position along the trajectory. Agreement rises monotonically from 41.2% in the first 5% of a trajectory to $\sim$90% past the 70% mark (pooled 76.0%); the probe's own correctness (last column) rises in lockstep (17.6% $\rightarrow$ 81.8%). Early probe reads are thus unstable to how the question is asked *and* unreliable as answers—the measurement side of the stability–terminality gap. The headline $\sim$57%/11% early/late disagreement contrast (Section 5.2) brackets these per-position values.

Takeaway. These observations—local consensus unreliable, recovery common, late consensus worse, stopped-but-wrong answers mostly *non-terminal* placeholders rather than settled mistakes, and even the probed answer being wording-unstable early—suggest that a good stopping rule must be much more than “agreement.” The rest of the paper asks whether the *best possible* such rule, drawn from a large preregistered space, can be both safe and economical. In the searched space, it cannot.